

Idea Validation

Adaptive Fitness Intelligence — National AI Challenge

Prepared by Manus AI · 20 August 2026

1 Verdict

Proceed to a tightly scoped MVP and evidence-generating pilot. The problem is real and technically addressable, but the proposition is not yet validated as a paid, retained, cross-platform service. Its strongest angle is explainable training context—not a generic AI readiness score.

1.1 1. Problem discovery

Wearable-derived [Heart-rate variability \(HRV\)](#) and [Resting heart rate \(RHR\)](#) are useful inputs but cannot fully prescribe a resistance-training session. In a 2024 validation study of 39 healthy adults with 316 paired recordings, Apple Watch HRV differed from the Polar H10/Kubios reference by −8.31 ms on average, whereas RHR differed by −0.08 bpm. [1] The product should therefore privilege repeated, within-device trends and combine them with workout-level data rather than impose universal thresholds. The structured strength log is scientifically relevant. A review covering 90 studies reported strong associations between resistance-training performance and practical signals including sessional, [Rate of perceived exertion \(RPE\)](#)- and [Repetitions in reserve \(RIR\)](#)-based measures, as well as selected velocity and force measures. [2] That supports the proposition that the user’s “load × reps × RPE” notes contain material decision context. Yet a 2022 meta-analysis across 15 studies found no statistically significant overall strength advantage for autoregulated versus standardised load prescription, so a coaching claim would be premature. [3]

1.2 2. Market reality check

Ireland is a credible launch laboratory rather than a huge standalone endpoint. The 2022 Census recorded 5,149,139 people, and Sport Ireland reported that 47% of adults—1.97 million people—regularly participated in sport in 2023. [4] [5] The Healthy Ireland Survey 2024 also found that 67% of people not meeting activity guidelines would like to be more active. [6] There is clear category infrastructure. IDC forecast 537.9 million worldwide wearable shipments in 2024; a U.S. survey of 23,974 adults found 44.5% reported wearable ownership in 2020–2022. [7] [8] Existing products validate the component jobs: WHOOP has recovery/workout data scopes; Cronometer syncs nutrition with devices; Strong and Hevy target lifting logs and progression. [9] [10] [11] [12] The gap is inferred from these scopes; it remains a hypothesis to test directly with customers.

1.3 3. Market-sizing scenario

Layer	Definition	Users	Revenue at €99/ year
Ireland Total addressable market (TAM)	Regular sports participants; theoretical ceiling.	1.97m	€195.0m
Ireland Serviceable addressable market (SAM)	15% power-user filter: regular lifters/endurance cross-trainers	295,500	€29.3m

Layer	Definition	Users	Revenue at €99/ year
	with wearable and detailed logging intent; assumption.		
Ireland early adopters	1% of SAM; realistic multi-year early adoption target, not a first-pilot target.	2,955	€292.5k ARR
Wider English-speaking scenario	US, UK, Canada, Australia and New Zealand population; applies U.S. wearable-ownership and Ireland active-sport proxies, then the same 15% filter.	15.15m	€1.50bn

The model is reproducible and intentionally conservative in presentation: €99 is a pricing hypothesis between Hevy Pro’s \$23.99 annual tier and WHOOP One’s \$199 annual EU/UK tier. [12] [13] Population inputs use World Bank 2024 data where available. The wider-market result is a directional sensitivity—not a revenue forecast—because country-specific wearable ownership, strength-training participation, consent rate and willingness to pay were not established in this review.

1.4 4. Scorecard

Criterion	Score	Justification
Problem clarity	8/10	Fragmentation and missing lifting context are concrete workflow problems with a clear decision moment.
Market demand	7/10	Device scale, sport participation and comparable products are positive signals; direct combined-product willingness to pay remains unproven.
Differentiation	7/10	Goal-aware strength semantics, ghost sets and explainable recommendations are a meaningful wedge, though components are copyable.
Monetisation	6/10	Comparable subscriptions support price discovery, but retention and standalone value are untested.
Scalability	6/10	A canonical data model scales, while private APIs, consent and normalization create friction.
Execution feasibility	5/10	Logging and insights are feasible; integrations, governance and prospective validation materially increase complexity.

Criterion	Score	Justification
Total	39/60	6.5/10: fundable as a disciplined MVP and research programme, not yet a proven platform-scale business.

1.5 5. Constraints that change the roadmap

WHOOP offers OAuth read scopes; Garmin's programme is business-oriented; Apple HealthKit is permissioned and users can revoke access or delete data outside the app. [9] [14] [15] MyFitnessPal's public information describes an approved/private API and Premium CSV export, so nutrition import must not depend on a presumed self-serve integration. [16] [17]

Data governance is a product feature. Ireland's Data Protection Commission treats health data as a special category and highlights explicit consent where applicable, transparency and data minimisation. [18] The European Commission describes the AI Act as a risk-based framework and emphasises transparency for AI use. [19] The MVP should offer granular consent, a recommendation audit trail and export/deletion controls, and avoid diagnostics or automatic medical instruction.

2 Glossary

HRV – Heart-rate variability: Variation in successive heartbeat intervals; treat as a device- and protocol-dependent trend input, not a diagnosis.

RHR – Resting heart rate: A repeated resting pulse measure that can be analysed as a personal trend.

RIR – Repetitions in reserve: Estimated repetitions remaining before technical failure.

RPE – Rate of perceived exertion: A subjective rating of effort for a set or session.

SAM – Serviceable addressable market: The reachable subset under explicit product, segment and geography assumptions.

TAM – Total addressable market: A theoretical category ceiling, not a sales forecast.

3 References

1. O'Grady et al. (2024), Apple Watch HRV/RHR validation.
2. Helms et al. (2020), resistance-training monitoring review.
3. Systematic review/meta-analysis (2022), autoregulation, strength and hypertrophy.
4. Central Statistics Office Ireland, Census 2022 Summary Results.
5. Sport Ireland, Irish Sports Monitor 2023.
6. Healthy Ireland Survey 2024.
7. IDC wearable shipment forecast (2024).
8. JMIR wearable ownership survey (2024).
9. WHOOP Developer API.
10. Cronometer Sync Devices.
11. Strong Workout Tracker.
12. Hevy Pricing.
13. WHOOP Membership Pricing.

14. Garmin Connect Developer Program.
15. Apple HealthKit Documentation.
16. MyFitnessPal developer API page.
17. MyFitnessPal data export support page.
18. Irish Data Protection Commission health-data guidance.
19. European Commission AI Act overview.